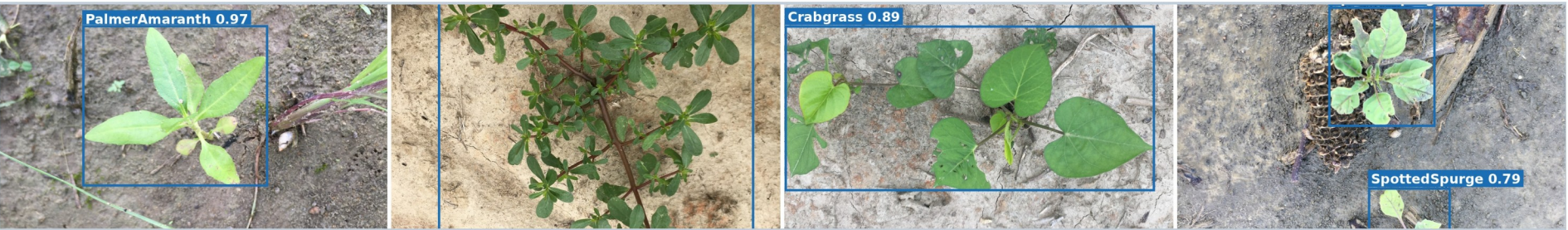


Every robot the MTSU Great Robotics Lab owns reports into one platform. Agents on it harvest, filter, label and train; a third watches those two. This is what six months of running it taught us.

WHAT EACH PROJECT IS, AND WHAT IT IS FOR

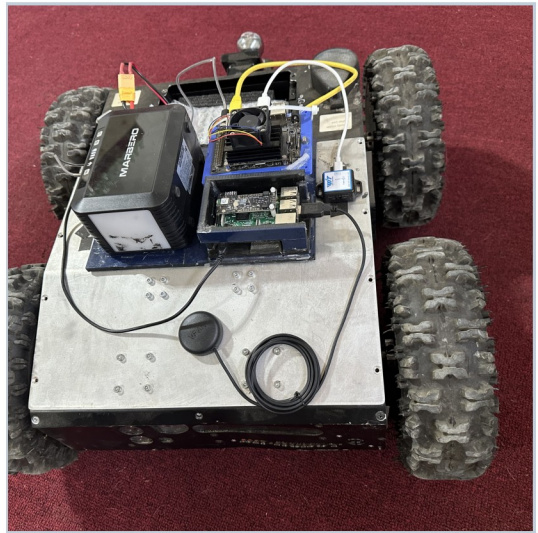
One platform, any dataset, any robot. Each capability stays in the platform after the experiment that needed it, so the next student inherits it working.

Weed Detection



Two agents work the corpus unasked: one harvests imagery and audits it, the other trains a detector small enough to ride the robots. More than 150,000 labelled images.

Rover



A tracked rover narrow enough to turn inside a row and reach where a tractor cannot. Camera, GPS and two inertial units stream live; the stream trains a driving model.

Laser Weed Control Robot



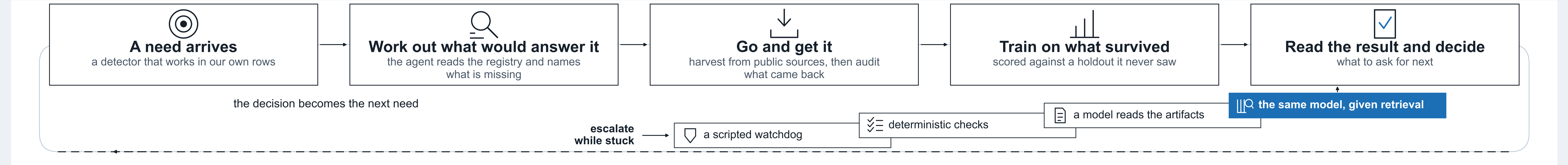
A twin-laser weeder. A YOLO detector finds the weed; two galvanometer lasers aim and fire, and it dies where it stands. No chemical, nothing touches the crop beside it.

Humanoid and Quadruped



A humanoid and a quadruped, out on the farm to record where wheels do not go. Their project takes the ingest contract the Rover uses; any session they open is governed.

Figure 1. Every project on the platform opened the same way, same ingest contract, same agents train and evaluate it.



Agent Work Flow Diagram

Figure 2. A need arrives, the agent works out what data would answer it, fetches it, trains on what survives the audit, scores, decides what to ask next. Stuck, it climbs the ladder. Dashed: the decision is recorded, nothing reads it back.

ONE PLATFORM TAKES ANY ROBOT, ANY DATASET

THE BRAIN GETS STUCK AND ASKS UPWARD

AGENTS RAN THE LOOP, NOBODY IN THE ROOM

MORE DATA STOPPED HELPING (CONT.)

- agentAI is one platform for every lab robot
- A robot, drone or sensor rig is a project with its own keys, people and locked dataset registry. Agents harvest, filter, label and train against it. Adding a robot is a configuration change.
- The Rover and the laser robot both report in
- The Rover runs on tracks and streams camera, GPS and two IMUs. The Laser Weed Control Robot carries a galvanometer laser, a down camera over the work zone and a forward camera.
- A drive becomes a dataset as it opens
- The platform registers a vehicle's session as a governed dataset while the drive runs. Frames and sensor batches arrive over one API key. The laser robot sends no position.
- Any data type, any field, one project each
- Open a project, upload, and get an automatic diagnosis before choosing what to do next. Weed detection is one project here; so are the Rover and the laser robot.

Grounded recall separates the reviewers

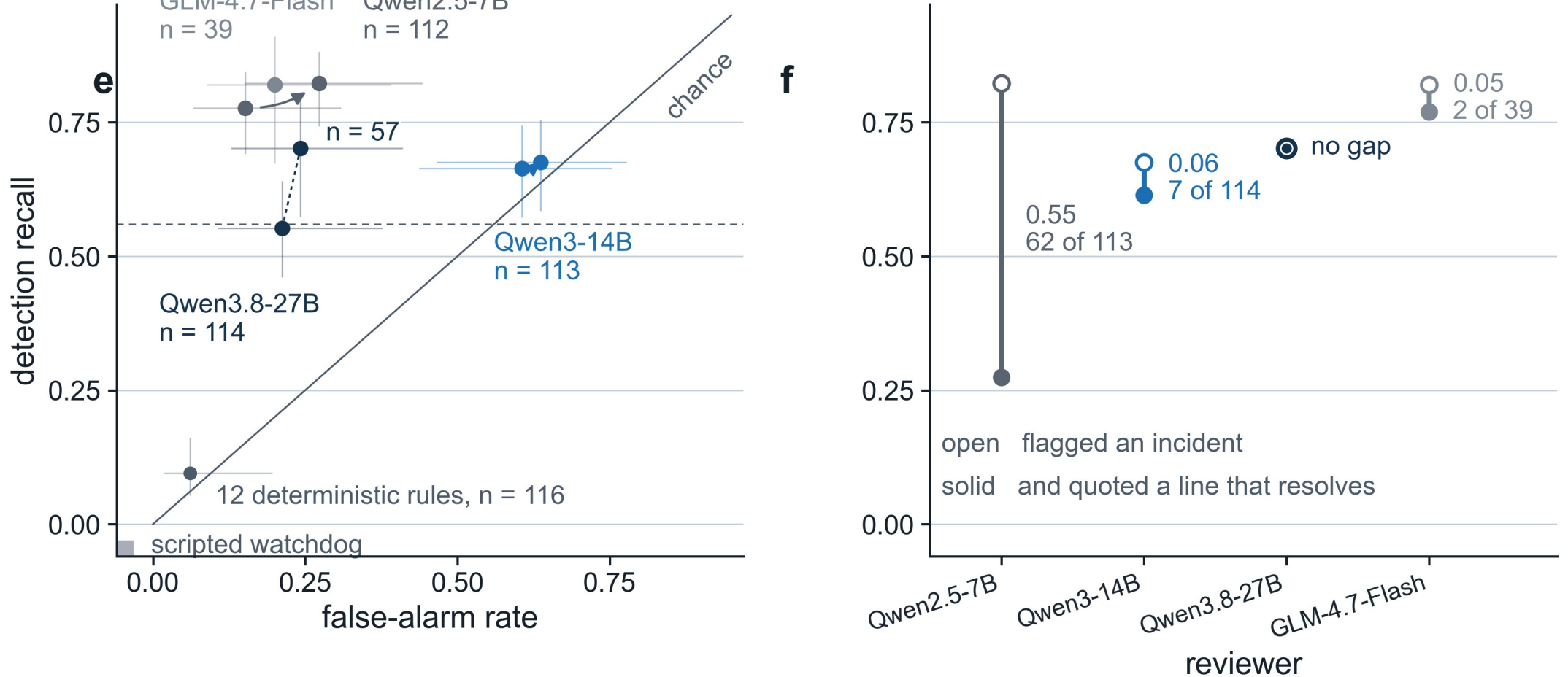


Figure 3. (e) Recall and (f) grounded recall on one frozen corpus of real incidents from our own record, scored by our own scorer. Tail: model on raw artifacts; head: same model with retrieval. Dotted arrow: ends scored on different cases.

Escalating is worth it, and the reason is grounding rather than size: retrieval is worth +0.149 recall to the 27 B and +0.012 to the 14 B. This part is built and is switched off today.

A cheap reviewer, and a bigger one above it when it is stuck

Review runs in tiers: a scripted watchdog, twelve deterministic checks, a model reading the raw artifacts, then the same model given retrieval. On 149 frozen cases the scripted watchdog returned no decision at all.

Fifteen rounds ran with nobody in the room

Two cluster jobs, one collecting and one training, ran against the locked registry. Every round reported success. This part is built and is switched off today.

MORE DATA STOPPED HELPING

More harvested data cost accuracy; we measured it

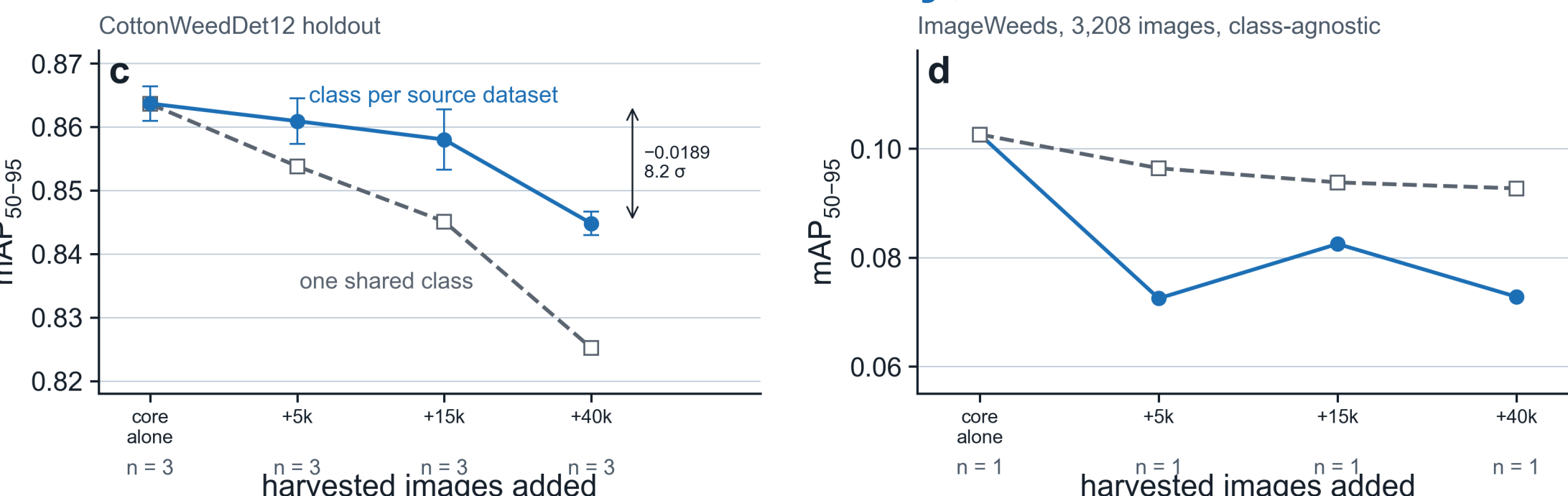


Figure 4. (c) Three seeds per rung, mean and 1 s.d., on our sealed holdout. (d) Same eight checkpoints, unseen weed dataset, one seed per rung. Both exams fall.

Starting fresh instead of from the last round's weights recovers +0.0287, 5.1 times this recipe's seed spread. That is the change the next campaign makes.

- The recipe was not why more data stopped helping
- More data arrived; the detector did not improve. Augmentation, replaying known species, completing labels: each left the same gap. What stood was our auto-labeller's boxes.
- A vision model can judge a dataset; a chat model cannot
- Professor Zhang's rule: discard what sits too far from data we trust. Shown a candidate box, Gemma 4 rejects none of the false ones, so the test is DINOv2 geometry. Fifty-one datasets out, retrained, accuracy did not come back.
- The unit of curation has to match the unit of noise
- An auto-label error is one wrong box in an otherwise fine image, so a dataset-level filter cannot see it. The per-box tools are written, not re-measured. That is the open front.
- A brain's falsifiable hypothesis, and it was wrong
- Harvested boxes carry a class taken from the website they came from. A brain blamed that, wrote its falsifying condition first, and a human approved. Four runs falsified it. The real cost was density: nine boxes an image against under two.

Paused since 2026-08-29. A stop-loss paused the collector and trainer. The Rover is off the tailnet. The supervisor is advisory. None of the 2,686 frames is labelled.

Every mAP is a best epoch. Most in-domain numbers are mAP50-95 on the same 1,977 images. That holdout is also the validation set during training, so every mAP here is a run's best epoch.

Numbers we withdrew. 0.910 was a data leak. 0.9033 is the best of four unseeded runs; only one of the four crossed 0.90. Reading the ladder as flat is retracted.